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Major Ion Hydrogeochemistry of Vadamalpet Mandal, Chittoor District, Andhra Pradesh

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Abstract

Major ion hydrogeochemistry of Vadamalapet Mandal is taken up for study and evaluation of its suitability for drinking and irrigation purposes. The study area comprising an area of 84 km^2 lies in between Latitudes 13°27' to 13°36' N and Longitude 79°26' to 79°34' E, and mainly occupied by granite, which is intruded by delerite dykes and quartz veins, quartzite with shaly intercalations, and alluvium. The paper presents the results of chemical analyses of 39 groundwater samples which are covering the entire sutdy area. The data has been tested for possible hazards due to total dissolved solids, sodium, bicarbonate, chloride, and sulphate. From the assessment, it is known that the majority of the water samples are good either for drinking or for agriculture in the study area.

Keywords: Major ion hydrogeochemistry, Groundwater, Suitability, Drinking, Irrigation.

INTRODUCTION

Hydrochemistry is an important aspect of groundwater in order to know its use for domestic, irrigation, and industrial purposes. The main factors that control the quality of groundwater are the associated lithology and soil. In addition, biological and climatic factors, and sources of pollution and other related aspects bear a direct influence on the chemical quality of groundwater. Analyses of water and its interpretation and assessment will determine the various uses of groundwater.

Vadamalapet mandal is an administrative unit in Chittoor district of Rayalaseema, Andhra Pradesh. The mandal lies in between Latitudes 13°27' to 13°36' N and Longitude 79°26' to 79°34'E and comprises an area of 84 km^2.

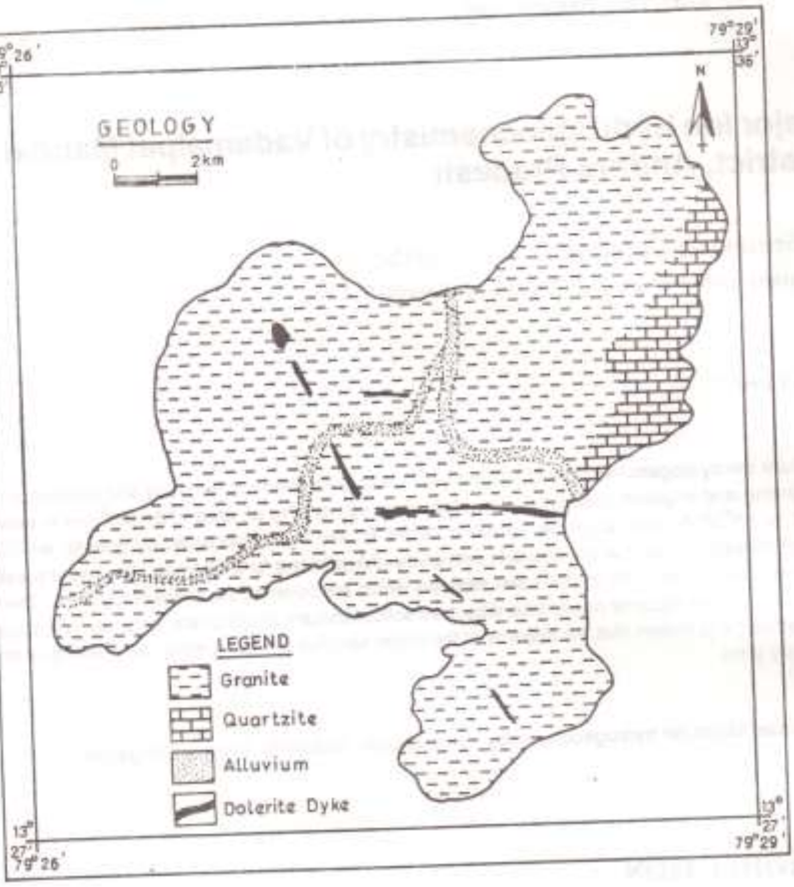

Fig. 1: Geological map of the study area.

GEOLOGY

The major portion of the study area is composed of Archean granite. Mineralogically it consist of quartz (SiO_2), potash feldspar ($KAlSi_3O_8$), sodic plagioclase ($NaAlSi_3O_8$), biotite $K(Mg,Fe)_3$ $(AlSi_3)O_{10}$ $(OH,F)_2$ and hornblende $(Ca,Mg,Fe,Na,Al)_{7-8}$ $(Al,Si)_8$ O_{22} in varying proportions. Microscopically it is coarse grained, equigranular to porphyritic. Dolerite dyke rocks are seen as black linear ridges running accross granitic hills and plain lands. They are almost vertical in many places and slightly inclined or near vertical in a few places and trend in all possible directions. More

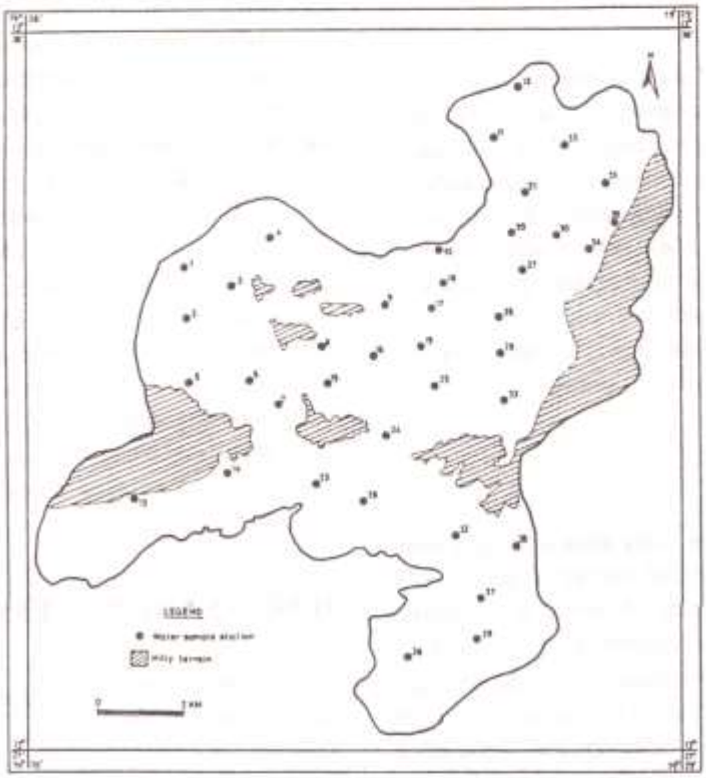

Fig. 2: Water samples location map.

commonly they trend E-W, NE-SW, NW-SE. Their relative age is indicated i.e., NS dykes are younger when they traverse the EW dykes. Essential minerals of dyke rocks include pyroxenes and plagioclases and the accessories are hornblende, biotite, sphene etc. Texturally they are ophitic to sub-ophitic though porphyritic texture is unknown. Next to it is quartzite with shally intercalations which belongs to lower cuddapahs of upper proterozoic age. These are Nagari quartzites and occur as outliers in granitic rocks trending northsouth exposing vertical cliffs forming escarpment. These rocks are fine grained and homogeneous in mineral composition. Rounded and sorted detrital quartz grains of quartzite are cemented by siliceous and ferrugenous material. Both vertical and horizontal joints are seen in all possible directions. Alluvium of recent age occurs along the main streams. It consists of boulders, sand, silt, and clay and ranges in thickness from the surface upto a depth of about 7.0 m (Fig.1).

Soils

The study area is composed of red, alluvial, alkaline and calcareous soils, Red soils are widely distributed over plain lands and are used for cultivation. Alluvial soils are less and occur only along the streams. Alkaline soils enriched with sodium carbonate occur as isoloated patches with random distribution in the study area. These soils neither allow growth of vegetation nor contain good quality of water.

Climate

Climate of the study area is hot and semi arid, with an annual rainfall of about 1001 mm. The intensity of rainfall is increases towards east of Vadamalapet. The study area often experiences both flood damage due to intense spell of rainfall as a result of storms and drought damage due to low or nil precipitation. It is observed that May is the hottest month of the every year with a monthly mean maximum of 39.14°C, a mean minimum of 26.55°C, and a mean temperature of 32.4°C.

METHODS OF STUDY

Groundwater samples were preserved in polythene bottles after collecting them normally below one meter depth from the general groundwater level. The constituents and properties of groundwater samples analysed include silica, calcium, magnesium, sodium, potassium carbonate, bicarbonate, sulphate, chloride, dissolved solids, hardness, alkalinity, non carbonate hardness, specific conductance (SEC), pH, sodium adsorption ratio, potential salinity, percent sodium, and residual sodium carbonate. Methods of collection and chemical analyses of water samples analysed and interpreterd are those of the methods given by Rainwater and Thatcher (1960), Brown et al. (1970), Hem (1970) and American Water Works Association (1971). Analyses for iron, manganese, and fluoride in a few samples indicate their occurrence in such small quantities that no attempt has been made to analyse them in the other samples.

RESULTS AND DISCUSSION

Thirty nine representative groundwater samples were collected from the dug wells, covering the entire study area as shown in Fig. 2. The maximum, minimum, and mean concentrations of different constituents of the water samples are given in Table 1.

The hydrogen ion concentration of the water samples ranges from 7.1 to 8.6 with a mean value of 7.85. About 60% of the water samples contain carbonate ion with pH is more than 8.3. Non carbonate hardness is present in about 50% of the water samples indicating that the total hardness exceeding carbonate hardness.

Then specific conductance versus dissolved solilds of the water samples was plotted on an arithmetic graph paper, the correlation is found to be positive (Fig. 3.)

Table 1: Chemical analyses of water samples.

S. No.	Constituents	Units	Min.	Max.	Mean
1	Silica	mg/l	1	12	5.69
2	Calcium	mg/l	11	157	56.23
3	Magnesium	mg/l	8	142	41.95
4	Sodium	mg/l	13	214	77.20
5	Potassium	mg/l	1	14	3.43
6	Carbonate	mg/l	8	43	22.27
7	Bicarbonate	mg/l	32	568	309.56
8	Sulphate	mg/l	2	100	30.54
9	Chloride	mg/l	21	653	119.26
10	Total disolved Solids	mg/l	143	1651	695.66
11	Hardness as $CaCO_3$	mg/l	72	976	307.87
12	Alkalinity as $CaCO_3$	mg/l	26	465	272.62
13	Non-carbonate Hardness	mg/l	3	563	99.46
14	Specific conductance	Micromhos/cm at 25°C	253	2720	1214.10
15	p^H		7.1	8.6	8.00
16	Sodium Adsorption Ratio		0.66	4.51	3.89
17	Percent Sodium		16	56	38
18	Percent Salinity		0.61	18.47	4.24
19	Residual sodium carbonate		0.009	2.19	0.42

The values of specific conductance are used to estimate approximately the total dissolved solids of any region by working out a factor. According to Hem (1970) the multiplying factor commonly varies from 0.54 to 0.96. The computed factor for the study area is 0.60.

Sodium Adsorption Ratio

The U.S. Salinity Laboratory (1954) has computed a ratio known as sodium adsorption ratio for determining the quality of water for agriculture which is given by a equation:

$$\text{Sodium Adsorption Ratio} = \frac{Na}{\sqrt{(Ca^{++})+(Mg^{++})/2}}$$

Irrigation waters are rated based on the classification given by Richards (1954). As the sodium asorption ratio values of the water samples in the study area are below 10, they belong to excellent category for irrigation. According to U.S. Salinity Laboratory

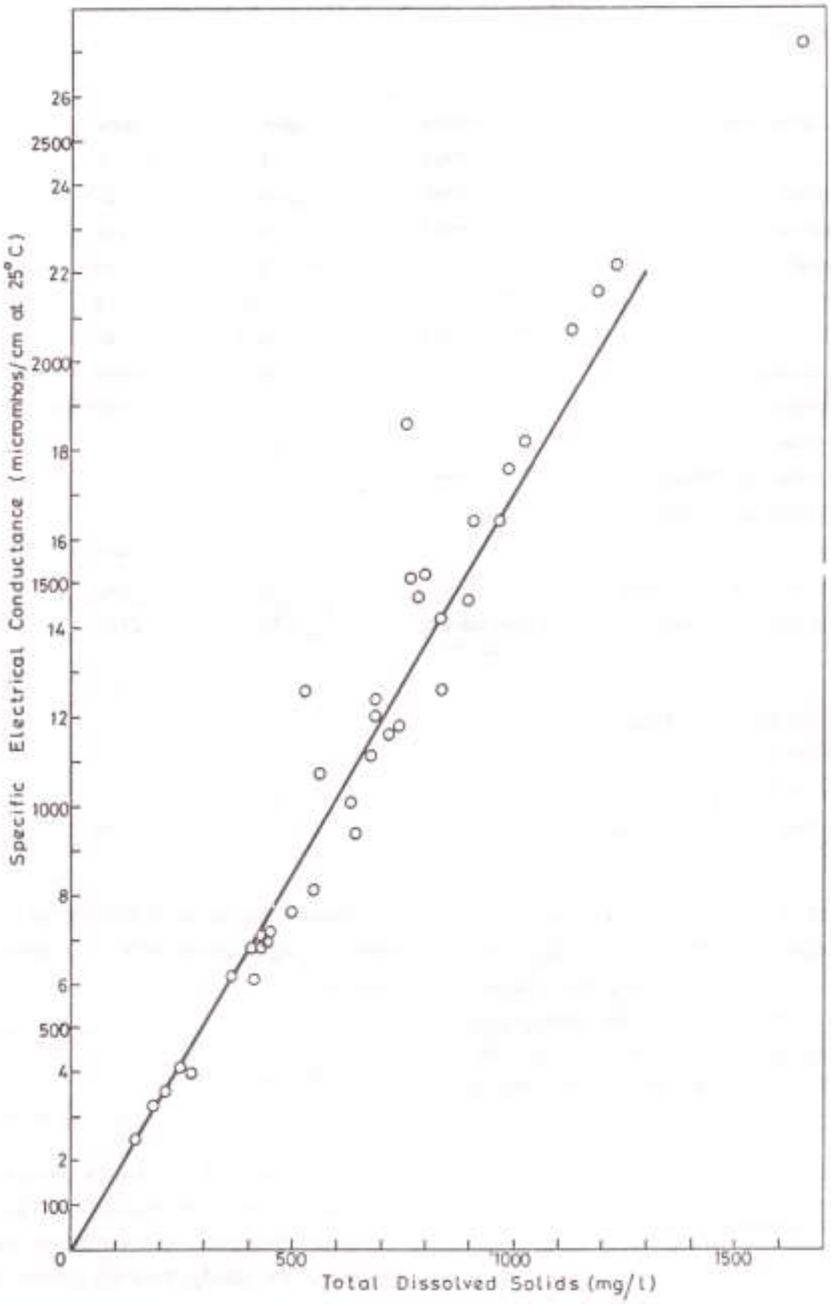

Fig. 3: Relationship between TDS and SES.

classification (Fig. 4). 31% of the water samples fall in C2S1 category with low sodium value and moderate sallinity hazard, while 64% of the water samples fall under C3S1 category, with low sodium and high salinity hazard. One sample each falls iln C4S1 and C3S2 category accounting for 5%.

Percent Sodium

The concentration of sodium is important in classifyinig irrigation waters because it reacts with soil affecting permeability. Sodium saturated soils namely alkaline soils formed on account of excess sodium in soil with carbonate as predominant anion or saline soils formed due to excess presence of sodium with either chloride or sulphate as predominant anion stunt the growth of plants. Therefore, Wilcox (1948) used percent sodium and specific conductance in evaluating irrigation waters using Wilcox diagram as shown in Fig. 5. The precent sodium is obtained from the equation:

$$\text{Percent Sodium} = \frac{Na+K}{Ca+Mg+Na+K} \times 100$$

According to the above classification 31% of the water samples belong to excellent to good category, 59% of the samples belong to good to permissible category and 10% of samples fall under doubtful to unsuitable.

Residual Sodium Carbonate (RSC)

The quantity of bicarbonate and carbonate in excess of alkaliine earths also influence the suitability of water for irrigation purposes (Eaton, 1950; Richards, 1954). This excess is known as residual sodium carbonate (RSC) and is obtained by using the equation:

$$\text{Residual Sodium Carbonate} = (CO_3^{-} + HCO_3^{-}) - (Ca^{++} + Mg^{++})$$

Where the concentrations are expressed in milliequivalents per litre.

According to U.S. Salinity Laboraory (1954), a residual sodium carbonate of less than 1.25 meq/l in waters is probably safe for irrigation while the waters are unsuitable for irrigation with residual sodium carbonate value more than 2.5 meq/l. Waters containing values between the above two can be rated as of marginal quality. The following table gives the results and rating of waters based on residual sodium carbonate.

RSC meq/1	Irrigation class	No. of samples	Precetage (%)
< 1.25	Safe	33	85
1.25 - 2.50	Marginal	6	15
> 2.50	Unsuitable	-	-

Based on the above classification, in 85% of the water samples residual sodium carbonate is absent and therefore, they are in safe limit and 15% of the water samples are

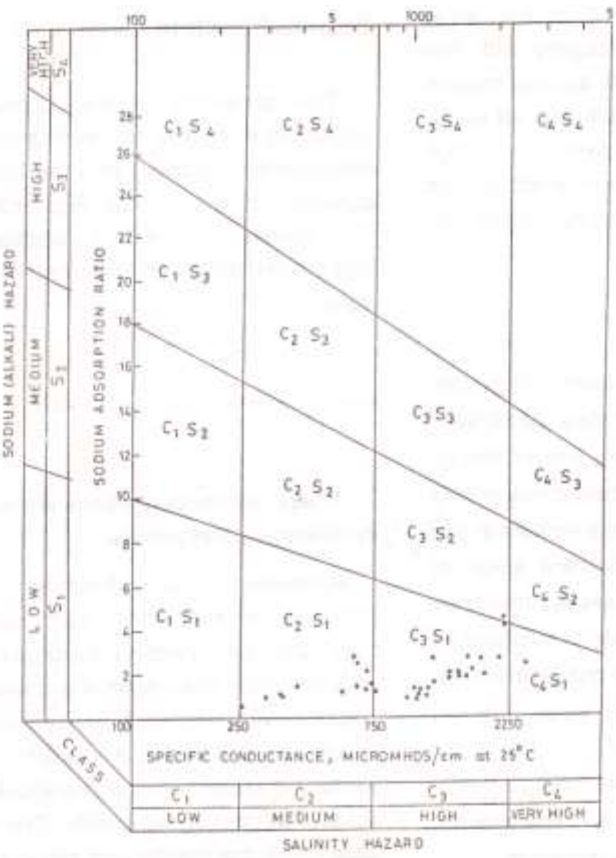

Fig. 4: U. S. Salinity diagram.

of marginal quality for irrigation. While none of them is unsuitable for irrigation.

Piper Diagram

Various trilinear methods of plotting are available namely Palmer (1911), Hill (1940) and Piper (1953). A modified trilinear method by Piper (1953) and Davis and De Wiest (1966) is widely used to understand the overall chemical character of the water samples in terms of cation and anion pairs. The diagram consists of three distinct fields viz. : two triangular fields and one intervening diamond shaped field whose sides are divided into 100 equal parts. The three cations, calcium, magnesium, and (sodium +

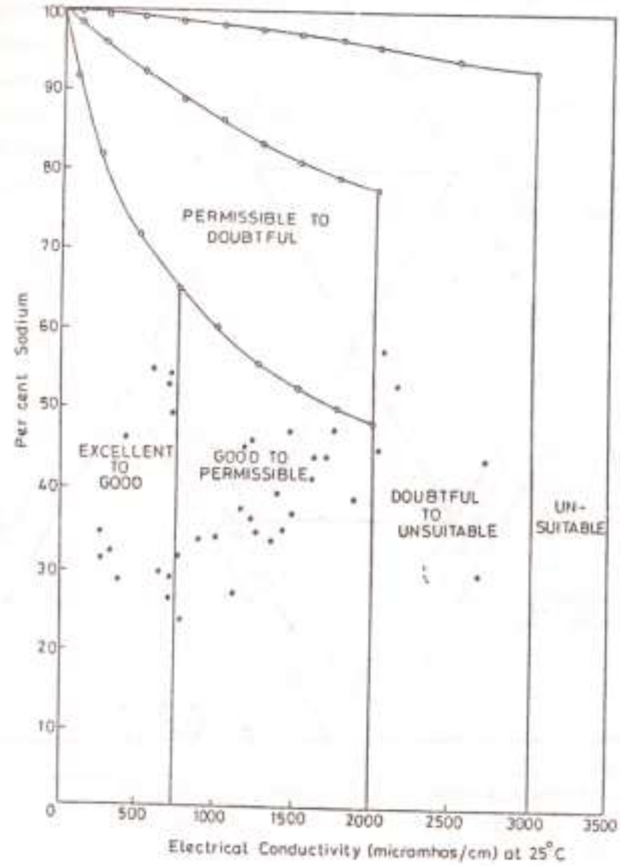

Fig. 5: Wilox (1984) diagram.

potassium), as percentage reacting values are plotted as a single point on the left triangular field while those of the anion group, bicarbonate+carbonate, chloride, and sulphate are similarly plotted on the right triangular field. The relative concentration of the several dissolved constituents of a water sample is shown by each point on the two triangular fields. The overall characteristics of the water are presented in the diamond shaped field by projecting the position of the plots in the two triangular fields as shown in Fig. 6.

Different types of groundwater can be distinguished by the position of their plottings occupying in the sub areas of the diamond shaped field. The groundwater of the study area is dominated by the alkaline eaths (Ca^{2+}

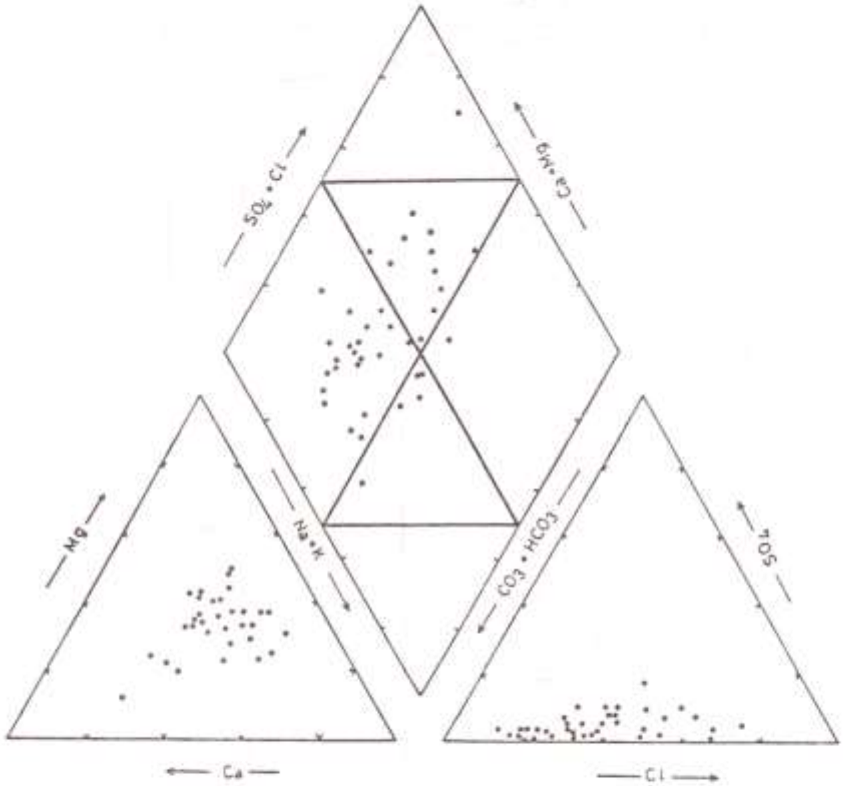

Fig. 6: Piper's trilinear diagram.

and Mg^{2+}) over the alkalies (Na^+ and K^+) and carbonate hardness (secondary alkalinity) exceeds to non carbonate hardness (secondary salinity).

Based on the Piper diagram, it is observed that 32 samples fall in area 1,7 samples fall in area 2, 26 samples fall in area 3, 13 samples in area 4, 21 samples in area 5, 1 sample in area6, 2 samples in area 7, 10 samples in area 9a and 5 samples fall in area 9b.

CONCLUSIONS

Assessment of groundwater samples from various methods proved that majority of the water samples are good either for drinking or for agriculture However, 15% of the samples are undesirable for drinking based on total dissolved solids, in about 64% of the samples, salinity hazard is found to be high according to US salinity laboratory diagram, and according to Wilcox diagram, about 10%

of the water samples fall under doubtful to unsuitable category. Accordinig to Piper's method of graphical representation of chemical analyses, the groundwater in the study area is dominated by the alkaline earths (Ca^{2+} and Mg^{2+}) over the alkalies (Na^{+} and K^{+}) and carbonate hardness (secondary alkalinity) exceeds to non carbonate hardness (secondary salinity). Continued use of the water of inferior quality is, perhaps, the cause for low agricultural returns. Application of gypsum treatment is recommended to waters with high sallinity, as they are harmful for irrigation purposes.

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